Soundness theorem for axiomatically presented modal logics

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof cites five validity facts from the sibling syntax-and-semantics unit (prop:valid-taut, prop:Kvalid, prop:subset-class, prop:soundMP, prop:Nec-rule), all of which are proved there; the prvDiamond conditional renders the non-diamond branch (tag inactive in this unit)

Editorial note. The label thm:soundness names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Validity of tautological and K instances. Tautological instances are valid in every class of models, and instances of the K axiom are valid in every class of models.. The induction basis handles length-1 proofs of the first two kinds by these validity facts.
- Soundness of MP and Necessitation. If C implies B and C are valid in a class of models then so is B, and if C is valid in a class then so is Box C.. The inductive step closes the modus ponens and Necessitation cases by these preservation facts.
- Class inclusion. If a formula is valid in a class C_i and C is a subclass of C_i, then it is valid in C.. The basis case passes from validity of an A_i instance in C_i to validity in the intersection class.

Source reference. Original proposition prop-Dual-valid: sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex, source lines 1480–1480. Named citation invoked at source line(s) 1480: the author's marker \olref[syn][sch]{prop:Dual-valid} resolves inside the full Open Logic Project to another unit (not staged

in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-Kvalid: sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex, source lines 1479–1479. Named citation invoked at source line(s) 1479: the author's marker \olref[syn][sch]{prop:Kvalid} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-soundMP: sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex, source lines 1490–1490. Named citation invoked at source line(s) 1490: the author's marker \olref[syn][sch]{prop:soundMP} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-valid-taut: sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex, source lines 1477–1477. Named citation invoked at source line(s) 1477: the author's marker \olref[syn][tau]{prop:valid-taut} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-Nec-rule: sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex, source lines 1493–1493. Named citation invoked at source line(s) 1493: the author's marker \olref[syn][val]{prop:Nec-rule} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-subset-class: sources/olp-normal-modal-logic-axioms-systems/pinned-original-source.tex, source lines 1483–1483. Named citation invoked at source line(s) 1483: the author's marker \olref[syn][val]{prop:subset-class} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Soundness Theorem)**.** *If every instance of A_1, ..., A_n is valid in the classes of models $[C]_1$, ..., $[C]_n$, respectively, then $\mathbf{K}A_1 \dots A_n \vdash B$ implies that B is valid in the class of models $[C]_1 \cap \dots \cap [C]_n$.*

Proof. By induction on length of proofs. For brevity, put $[C] = [C]_1 \cap \dots \cap [C]_n$.

1. Induction Basis: If B has a proof of length 1, then it is either a tautological instance, an instance of K, or of d,or an instance of one of A_1, ..., A_n. In the first case, B is valid in $[C]$, since tautological instance are valid in *any* class of models, by prop-valid-taut. Similarly in the second case, by prop-Kvalid and prop-Dual-valid. Finally in the third case, since B is valid in $[C]_i$ and $[C] \subseteq [C]_i$, we have that B is valid in $[C]$ as well by prop-subset-class.

2. Inductive step: Suppose B has a proof of length $k > 1$. If B is a tautological instance or an instance of one of A_1, ..., A_n, we proceed as in the

previous step. So suppose B is obtained by MP from previous formulas $C \to B$ and C. Then $C \to B$ and C have proofs of length $< k$, and by inductive hypothesis they are valid in $[C]$. By prop-soundMP, B is valid in $[C]$ as well. Finally suppose B is obtained by NEC from C (so that $B = \Box C$). By inductive hypothesis, C is valid in $[C]$, and by prop-Nec-rule so is B.

□